

Chapitre 6 : Instructions itératives

Solution des exercices

Exercice 1

```
import java.util.Scanner;

public class Chap6Ex1 {

    public static void main(String[] args) {
        String texte;
        int nbRep;
        String reponse = "";
        Scanner sc = new Scanner(System.in);

        System.out.print("Texte ? ");
        texte = sc.nextLine();

        System.out.print("Nombre de répétitions ? ");
        nbRep = sc.nextInt();

        for(int i = 0; i < nbRep; i++){
            reponse += texte + "\n";
        }

        System.out.println(reponse);
    }
}
```

Exercice 2

```
public class Chap6Ex2 {

    public static void main(String[] args) {
        String suite = "";

        for(int i = 2; i <= 30; i = i + 2){
            suite = suite + " " + i;
        }

        System.out.println(suite);
    }
}
```

```
}  
}
```

Exercise 3

```
public class Chap6Ex3 {  
    public static void main(String[] args) {  
        String suite = "";  
  
        for(int i = 10; i <= 150; i+=10){  
            suite = suite + " " + i;  
        }  
  
        System.out.println(suite);  
    }  
}
```

Exercise 4

```
import java.util.Scanner;  
  
public class Chap6Ex4 {  
    public static void main(String[] args) {  
        int entier1;  
        int entier2;  
        int debut;  
        String suite = "";  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Premier entier ? ");  
        entier1 = sc.nextInt();  
  
        System.out.print("Dernier entier ? ");  
        entier2 = sc.nextInt();  
  
        debut = entier1;  
        if (debut % 2 != 0)
```

```

        debut++;

        for(int i = debut; i <= entier2; i += 2)
            suite += i + " ";

        System.out.println(suite);
    }
}

```

Exercise 5

```

import java.util.Scanner;

public class Chap6Ex5 {

    public static void main(String[] args) {
        int entier1;
        int entier2;
        int tmp;
        String suite = "";
        Scanner sc = new Scanner(System.in);

        System.out.print("Premier entier ? ");
        entier1 = sc.nextInt();

        System.out.print("Dernier entier ? ");
        entier2 = sc.nextInt();

        if (entier1 > entier2) {
            tmp = entier1;
            entier1 = entier2;
            entier2 = tmp;
        }

        for(int i = entier1; i <= entier2; i++)
            suite += i + " ";

        System.out.println(suite);
    }
}

```

Exercice 6

```
import java.util.Scanner;

public class Chap6Ex6 {

    public static void main(String[] args) {
        int entierDepart;
        int NombreMultiple;
        String suite = "";
        Scanner sc = new Scanner(System.in);

        System.out.print("Entier de départ ? ");
        entierDepart = sc.nextInt();

        System.out.print("Nombre de multiples ? ");
        NombreMultiple = sc.nextInt();

        for (int i = 1; i <= NombreMultiple; i++) {
            suite += (i * entierDepart) + " ";
        }

        System.out.println(suite);
    }
}
```

Exercice 7

```
import java.util.Calendar;
import java.util.GregorianCalendar;
import java.util.Scanner;

public class Chap6Ex7 {

    public static void main(String[] args) {
        int annee;
        int mois;
        int premierJourDuMois;
        int nombreJourDansLeMois;
```

```

String texteCalendrier = "Dim Lun Mar Mer Jeu Ven Sam\n";
Scanner sc = new Scanner(System.in);

System.out.print("Année ? ");
annee = sc.nextInt();

System.out.print("Mois ? ");
mois = sc.nextInt();

GregorianCalendar uneDate = new GregorianCalendar(annee, mois, 1);
premierJourDuMois = uneDate.get(Calendar.DAY_OF_WEEK);

if (mois == 1)
    if (uneDate.isLeapYear(annee))
        nombreJourDansLeMois = 29;
    else
        nombreJourDansLeMois = 28;
else
    if (mois == 0 || mois == 2 || mois == 4 || mois == 6 || mois == 7 || mois == 9 || mois
== 11)
        nombreJourDansLeMois = 31;
    else
        nombreJourDansLeMois = 30;

for(int i = 1; i < premierJourDuMois; i++)
    texteCalendrier += "    ";

for(int i = 1; i <= nombreJourDansLeMois; i++) {
    texteCalendrier += String.format("%02d ", i);
    if((i + premierJourDuMois - 1) % 7 == 0)
        texteCalendrier += "\n";
    else
        texteCalendrier += " ";
}

System.out.println(texteCalendrier);
}
}

```

Exercise 8

```

import java.util.Scanner;

public class Chap6Ex8 {

    public static void main(String[] args) {
        String texte;
        String reponse = "";
        Scanner sc = new Scanner(System.in);

        System.out.print("Texte ? ");
        texte = sc.nextLine();

        for(int i = 0; i < texte.length(); i++){
            if (i ==0)
                reponse += texte.charAt(i);
            else
                reponse += "-" + texte.charAt(i);
        }

        System.out.println(reponse);
    }
}

```

Exercise 9

```

import java.util.Scanner;

public class Chap6Ex9 {

    public static void main(String[] args) {
        String texte;
        String texteInv = "";
        Scanner sc = new Scanner(System.in);

        System.out.print("Texte ? ");
        texte = sc.nextLine();

        for (int i = 0; i < texte.length(); i++)
            texteInv = texte.charAt(i) + texteInv;

        if(texteInv.equalsIgnoreCase(texte))

```



```
        case 'ï' :   texteSansAccents += 'i';
                    break;
        case 'ô' :
        case 'ö' :   texteSansAccents += 'o';
                    break;
        case 'ù' :
        case 'û' :
        case 'ü' :   texteSansAccents += 'u';
                    break;
        case 'ç' :   texteSansAccents += 'c';
                    break;
        default :   texteSansAccents += texteSansEspaces.charAt(i);
    }

    for (int i = 0; i < texteSansAccents.length(); i++)
        texteInv = texteSansAccents.charAt(i) + texteInv;

    if(texteInv.equalsIgnoreCase(texteSansAccents))
        System.out.println "\"" + texte + "\" est un palindrome");
    else
        System.out.println "\"" + texte + "\" n'est pas un palindrome");
}
}
```

Exercice 11

```
import java.util.Locale;
import java.util.Scanner;

public class Chap6Ex11 {

    public static void main(String[] args) {
        double montant;
        double taux;
        int nbAnnee;
        String affichage = "Année\tValeur du placement\n";
        Scanner sc = new Scanner(System.in).useLocale(Locale.US);

        System.out.print("Montant du placement ? ");
        montant = sc.nextDouble();

        System.out.print("Taux d'intérêt annuel ? ");
        taux = sc.nextDouble();

        System.out.print("Nombre d'années du placement ?");
        nbAnnee = sc.nextInt();

        for (int i = 1; i <= nbAnnee; i++)
        {
            montant = montant * (1 + taux);
            affichage += "" + i + "\t" + String.format("%.2f", montant) + " $\n";
        }

        System.out.println(affichage);
    }
}
```